* Transport Layer

→ It is located between the application layer and the network layer.

→ It is a logical Communication between two processes.

→ On the Sender, the transport-layer
  - → receives messages from an application process
  - → Converts the messages into the Segments
  - → passes the Segment to the network layer.

→ On the receiver, the transport-layer
  - → receives the Segment from the network layer
  - → Converts the segments into to the messages
  - → Passes the messages to the application layer

→ The Internet has 2 Transport layer protocols
  1) TCP (Transmission Control Protocol)
  2) UDP (User Datagram Protocol)

1) UDP (User Datagram Protocol) :-

→ UDP provides a connectionless service to the invoking application

→ UDP Provides (i) Process-to-process data delivery
  (ii) Error checking

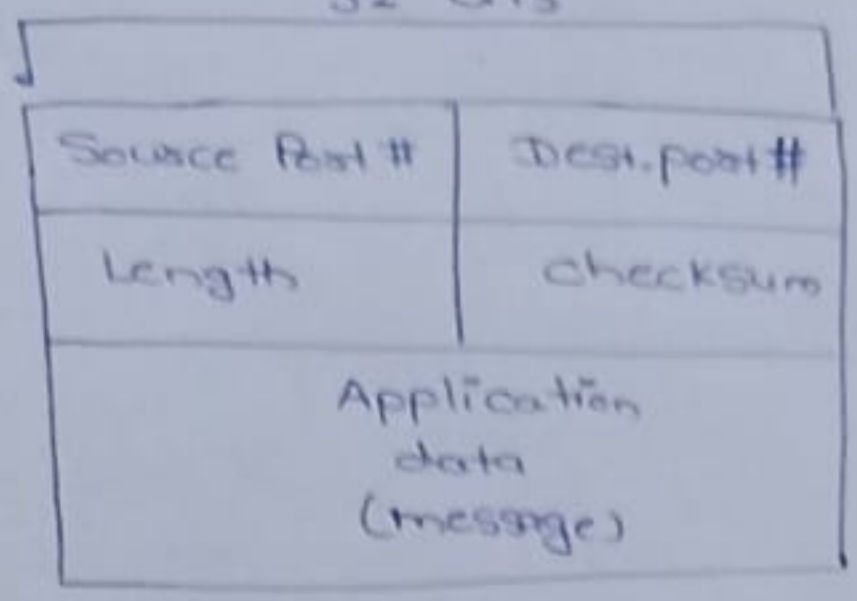
→ UDP is an unreliable service i.e, it doesn't guarantee data will arrive to destination-process

2) TCP (Transmission Control Protocol) :-

→ TCP provides a connection-oriented Service to the invoking application.

→ TCP provides (i) Reliable data transfer
  (ii) Congestion control
  (iii) Error checking

* UDP Segment Structure :-



→ UDP Segment Contains

1) Application Data :- This field occupies the data-field of the Segment.

2) Destination Port No :- This field is used to deliver the data to correct process running on the destination-host.

3) Length : This field Specifies the number of bytes in the Segment.

4) checksum : This field is used for error-detection

* Advantages of TCP over UDP :-

1) Reliability :- TCP ensures reliable data delivery by retransmitting lost or corrupted packets which UDP does not.

2) Data Integrity :- TCP guarantees data is delivered in correct order, maintaining the sequence of the transmitted packets.

3) Connection-oriented :- TCP establishes a connection before data transfer, ensuring both Senders & receivers are ready for communication.

4) Error checking :- TCP performs error detection and automatically recovers from errors by retransmitting affected packets.

5) Flow Control :- TCP uses flow control mechanism to prevent network congestion and adjust the data flow between Senders and receivers.

* TCP Connection Management :-

1) TCP three-way handshake :-

→ To setup the connection, three segments are sent between the two hosts. Therefore, this process is referred to as three-way handshake.

→ It establish a reliable connection between a client and a Server before data transmission begins

Step 1 : Client Sends a Connection request Segment to Server

→ The client first Sends a Connection request Segment to the Servers.

→ The Connection-request Segment Contains

1) SYN bit is Set to 1 (Synchronize)
2) Initial Sequence-number (Client_isn)

→ The SYN Segment is encapsulated within an IP datagram and Sent to the Servers

Step 2 : Server Sends a Connection granted Segment to Client

→ Server responds to the client request with SYN-ACK Signal bits Set.

→ Acknowledgement (ACK) Signifies the response of the Segment it received and SYN Signifies with What Sequence number is likely to Start the Segment with.

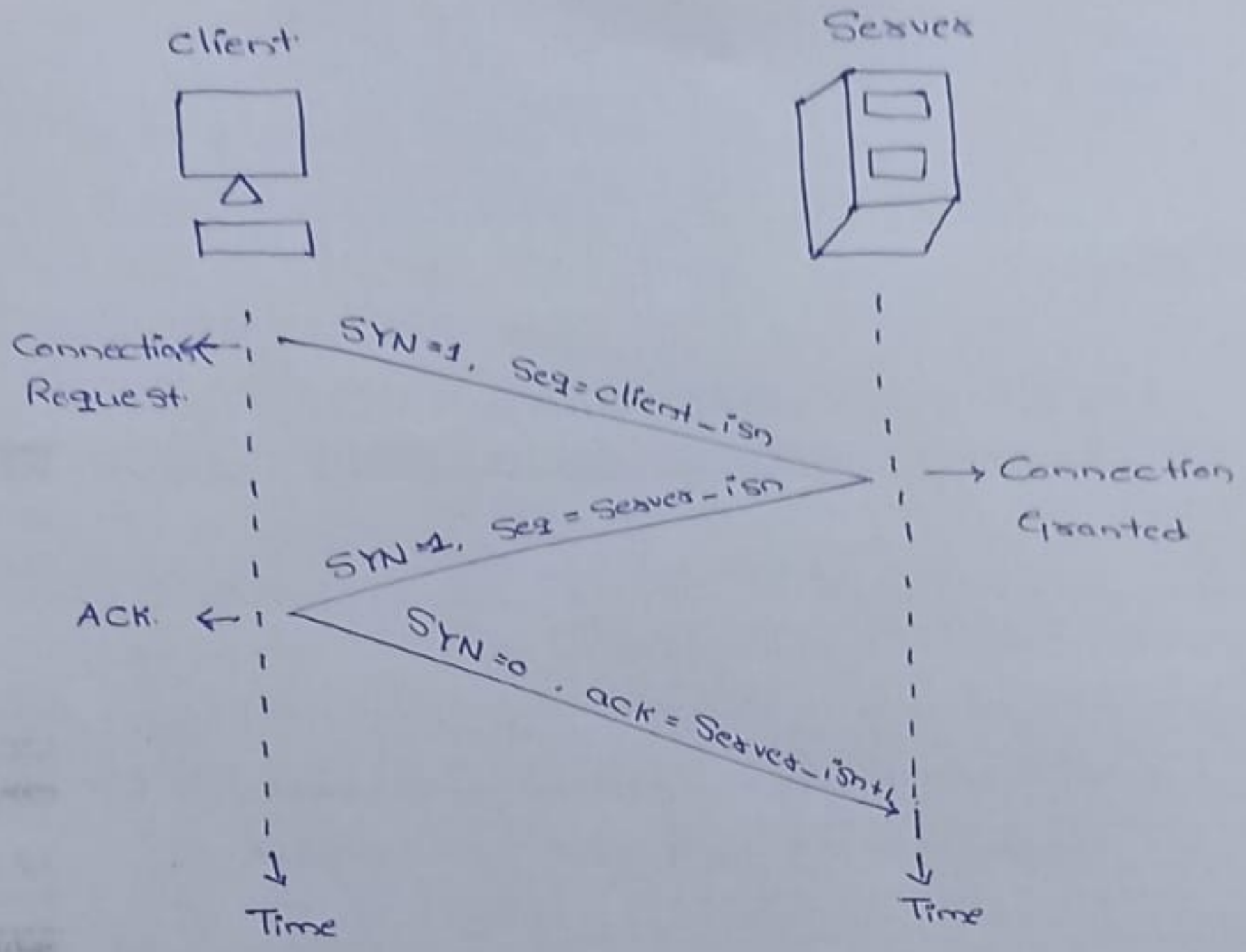
→ The connection-granted Segment Contains:

1) SYN bit is Set to 1.
2) Acknowledgement field is Set to client_isn+1.
3) Initial Sequence-number (Servers-isn).

Step 3 : Client Sends an ACK Segment to the Servers

→ In the final part client acknowledges the response of the Servers and they both establish a reliable Connection.

→ SYN bit is Set to Zero, Since the Connection is established.



* Client - Server Architecture :-

→ In this architecture, there is a server and many clients distributed over the network.

→ The Server is always-on, while a client can be randomly run.

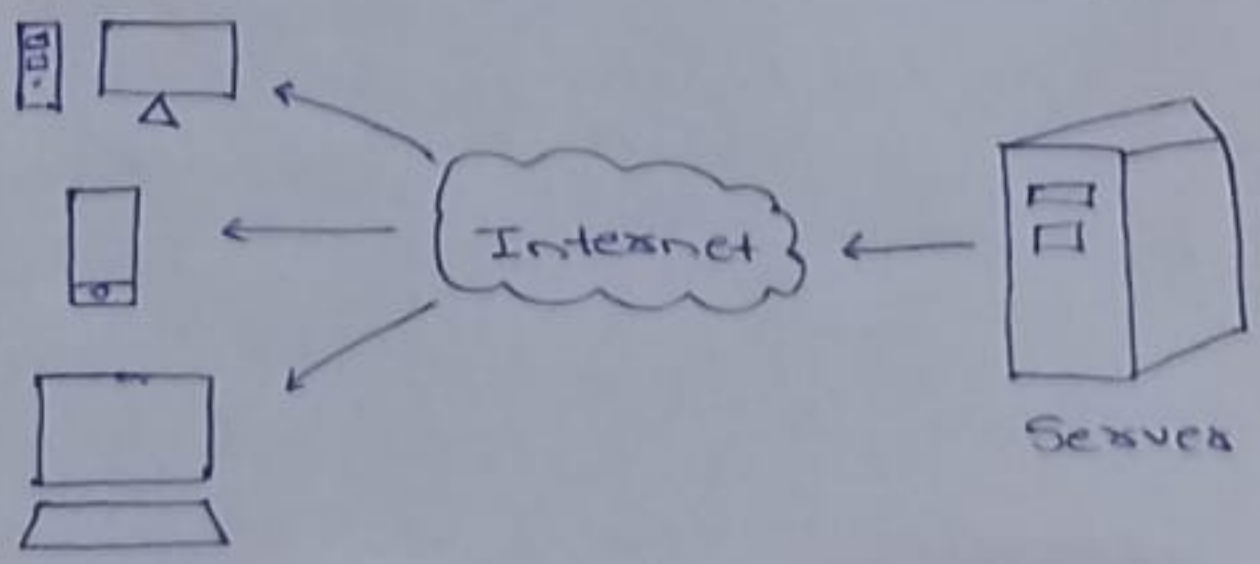
→ Upon requests from a client, the server provides certain services to the client.

→ Usually, there is no communication between two clients

→ The Server has a fixed IP address.

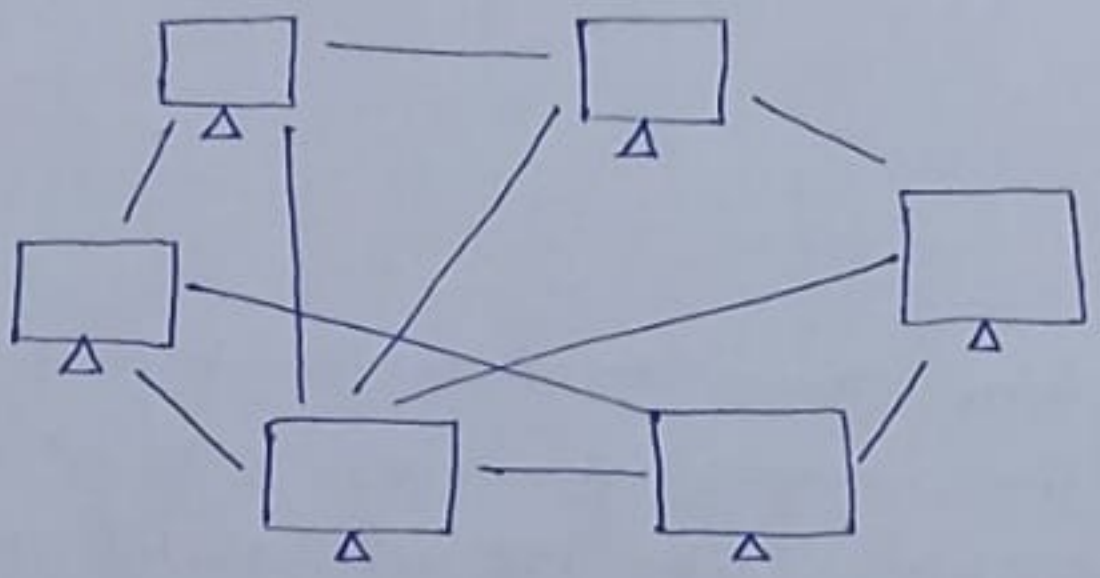
→ A client contacts the Server by sending a packet to the Server's IP address.

→ The applications such as FTP, telnet, Web, e-mail etc use the client - Server architecture



* Peer to peer Architecture :-

→ There is no dedicated Server.
→ pairs of hosts are called peers.
→ The peers communicate directly with each other
→ The peers are not owned by the Service provider, but are instead desktops and hptops controlled by users
→ Many of today's most popular and traffic - intensive applications are based on P2P architecture.
→ Ex: File Sharing (BitTorrent), Internet telephone (Skype)



* SMTP (Simple Mail Transfer Protocol) :-

→ SMTP is an application layer protocol used for email.
→ SMTP uses TCP to transfer mail from Sender's mail Server to the recipient mail Server.
→ Three characteristics of SMTP are.
  1) Message body uses 7-bit ASCII code only
  2) Normally, no intermediate mail Servers used for sending mail.
  3) Mail transmission across multiple networks through mail relaying.

→ How it works.
  1) Usually, mail-Servers are listening at port 25
  2) The Sending Server initiates a TCP Connection to the receiving mail-Server.
  3) If the receiver Server is down, the Sending Server will try later.

4) Then, the client indicates the e-mail address of the Sender and the Recipient.

5) Finally, the client Sends the message to the Server over the Same TCP Connection.

* DNS:

→ DNS is a System that translates human-readable domain names (like www.google.com) into machine readable IP addresses (like 198.105.232.4) which Computers can Communicate.

→ Key features of DNS

1) Distributed Database
2) Application Layer protocol
3) Uses UDP and Port 53.

→ How DNS Works:

1) Browser extracts the hostname from the URL
2) Browser passes the hostname to the DNS application on our machine
3) The DNS client Sends a query to the DNS Server
4) The DNS Server replies with the corresponding IP address.
5) The browser uses the IP address to establish a Connection to the website.